

Turbulence Modeling Resource

LES: 2-D NASA Wall-Mounted Hump 2009

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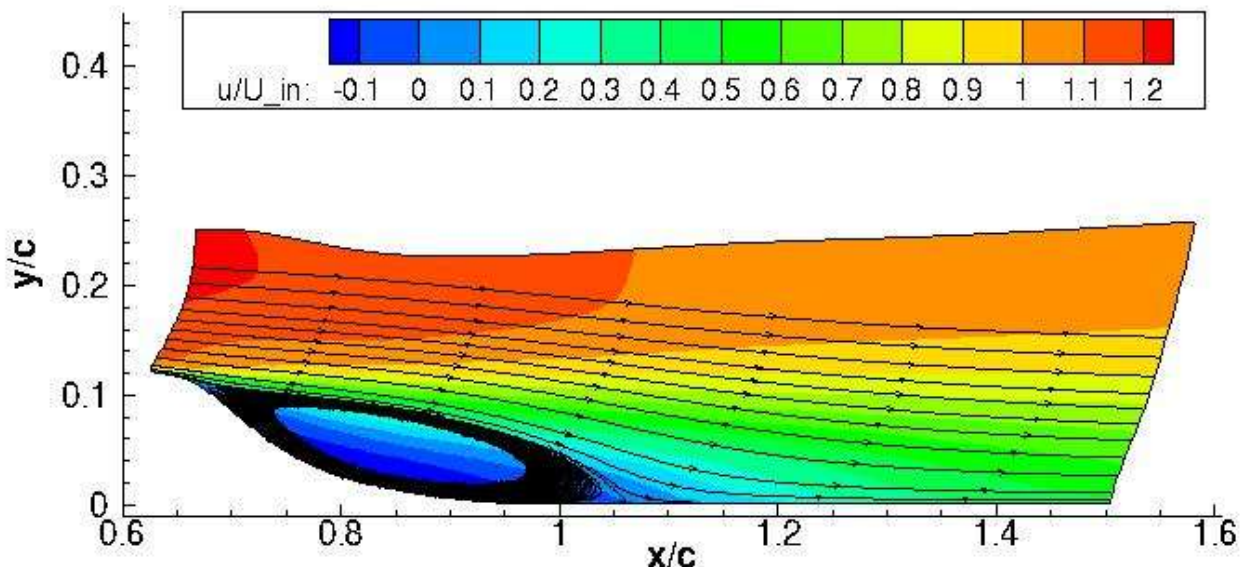
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The data on this page were provided by [S. Lardeau](#) of CD-ADAPCO.

This LES case is for 2-D separating flow over the NASA Wall-Mounted Hump (no flow control). The link to the experimental data can be found by clicking here: [NASA CFDVAL2004 Workshop page](#) (independent website, will open new window). The LES was run using an incompressible code, and has been documented in the following publication:

- Avdis, A., Lardeau, S., and Leschziner, M. A., 2009. "Large Eddy Simulation of Separated Flow over a Two-dimensional Hump with and without Control by Means of a Synthetic Slot-jet," Flow Turbulence Combust 83, pp. 343-370, <https://doi.org/10.1007/s10494-009-9218-y>.

Only limited LES data are provided (not all LES data from the 2009 publication are available).



Some relevant information is given here, but the interested reader is referred to the above publications for complete details:

- Reynolds number (based on U_{in} and c) = 935,892, where U_{in} is the freestream velocity and c is the hump chord.
- At $x/c = -2.14$, $Re_{\theta} = 6771$ and $B.L. \text{ thickness} = 0.07c$.

Long-time averaged LES data are provided below (on a grid normalized by the hump chord). These results are from a grid with approximately 9.4 million grid cells, covering a spanwise extent of 0.4 c with 128 cells in the spanwise direction. The 768 x 96 x 128 grid extended from $-0.81 < x/c < 2$, and the upper wall at $y/c=0.909$ included a constriction to mimic the blockage effect of the end plates in the experimental configuration. A precursor simulation with rescaling was performed in a region upstream and overlapping the hump grid, in order to establish the inflow conditions. Additional details about the LES grid and methodology can be found in the above reference. Details regarding the NASA wall-mounted hump geometry can be found at: [Separated Flow over Wall-mounted Hump with and without Flow Control](#).

The full grid used in the LES simulation is not available here. The field data below contains LES time-averaged results only on a limited region from the full grid (extracted in the area of interest shown in the figure above using every other point in each coordinate direction from the original LES grid).

Velocity and turbulence mean field data:

- [LES_meanfield_nasahump2009.dat \(limited region\)](#)
- [LES_meanfield_nasahump2009_tec.dat \(limited region\) - Tecplot readable](#)

Extracted velocity and turbulence profiles at $x/c=0.65, 0.66, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2$, and 1.3 :

- [LES_statistics_profiles_nasahump2009.dat](#)

Surface pressure coefficients:

- [LES_cp_nasahump2009.dat](#)

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